DSC Capstone LaTeX Template

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Abstract

Replace the `lipsum[1]` in the source code with your abstract. Don't include an abstract with your Quarter 2 Proposal. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Code: https://github.com/VedVar43789/na-rfm-pets

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1 Introduction

Diffusion models generate images by progressively transforming random noise into structured samples through a learned denoising process (Ho, Jain and Abbeel 2020). However, producing realistic images does not by itself provide control over what those images depict. A pretrained model may generate a broad range of objects while offering no direct way to request a particular category or visual attribute. Adding control after training lets us guide the model toward specific categories or attributes without retraining it. The central challenge is to guide generation toward a target concept while preserving image quality and keeping the computational cost manageable.

Our project studies this problem through Noise-Aligned Recursive Feature Machine Steering (NA-RFM), introduced by Wang, Belkin and Wang (2026). NA-RFM constructs guidance signals from labeled examples while keeping the pretrained diffusion model fixed. It combines a correction based on principal component analysis (PCA) at high noise levels with a learned direction in the model's internal activation space at intermediate and low noise levels. These signals are computed offline and reused during generation, avoiding inference-time backpropagation.

We aim to reproduce the paper's ImageNet steering experiment using the released implementation, then adapt the method to the Oxford-IIIT Pet dataset. This adaptation will compare two levels of semantic control: generating cats versus dogs, and generating individual breeds. We will also investigate whether an activation direction learned at one noise level remains effective when applied at other noise levels. Our evaluation will examine target-class accuracy, Fréchet Inception Distance (FID), and computational cost, with offline preparation reported separately from sampling. Together, these experiments will assess how the method's effectiveness changes with target granularity and with the transfer from ImageNet to a different image dataset.

2 Methods

3 Results

4 Discussion

5 Conclusion

References

Ho, Jonathan, Ajay Jain, and Pieter Abbeel. 2020. "Denoising Diffusion Probabilistic Models." In *Advances in Neural Information Processing Systems*. [Link]

Wang, Qingsong, Mikhail Belkin, and Yusu Wang. 2026. "General and Efficient Steering of Diffusion Models." *arXiv preprint arXiv:2602.11395*. [Link]